

River Restoration in Urban Areas

Questionnaire Survey + Choice Experiment

“The aim of this survey is to find out the attitudes and preferences of the inhabitants of this city towards possible modifications of the existing watercourse. All data provided by you will not be shared on an individual level and will be anonymized.”

What role does nature play in your life? For each of the following statements, **indicate how much you agree with them.**

- **Nature is an important part of my life.**

Strongly disagree	Rather disagree	Rather agree	Strongly agree	I can't judge
-------------------	-----------------	--------------	----------------	---------------

- **What I appreciate about nature is its diversity.**

Strongly disagree	Rather disagree	Rather agree	Strongly agree	I can't judge
-------------------	-----------------	--------------	----------------	---------------

- **The wilder the nature, the more I like it.**

Strongly disagree	Rather disagree	Rather agree	Strongly agree	I can't judge
-------------------	-----------------	--------------	----------------	---------------

How often do you spend your free time in green areas in general?

- Daily
- Several times a week
- Several times a month
- Less often

Where did you just come from?

- ☐ From home
- ☐ From work/university/school
- ☐ From shopping
- ☐ From kindergarten/nursery
- ☐ From somewhere else:

How do you usually get to this location?

- ☐ Walking
- ☐ By bicycle
- ☐ By public transport
- ☐ By car
- ☐ Combination 1: Walking + By public transport
- ☐ Combination 2: Walking + By car
- ☐ Other (e.g., inline skates, scooter, ...):

How often do you visit this location near “***NAME OF stream/river***”?

- ☐ Daily
- ☐ Several times a week
- ☐ Several times a month
- ☐ Less often
- ☐ This is my first time here

How much time do you **usually** spend here?

- ☐ I'm just walking through
- ☐ Up to 15 minutes
- ☐ 15-30 minutes
- ☐ 31-60 minutes
- ☐ More than 1 hour

What is your reason for visiting this area? For each of the following statements, **indicate how much you agree with them.**

I want to...

- *... get from "A to B"*

Strongly disagree	Rather disagree	Rather agree	Strongly agree	I can't judge
-------------------	-----------------	--------------	----------------	---------------

- *... be active in sports*

Strongly disagree	Rather disagree	Rather agree	Strongly agree	I can't judge
-------------------	-----------------	--------------	----------------	---------------

- *... talk and be with other people*

Strongly disagree	Rather disagree	Rather agree	Strongly agree	I can't judge
-------------------	-----------------	--------------	----------------	---------------

- *... enjoy some peace, relax*

Strongly disagree	Rather disagree	Rather agree	Strongly agree	I can't judge
-------------------	-----------------	--------------	----------------	---------------

- *... observe and experience nature*

Strongly disagree	Rather disagree	Rather agree	Strongly agree	I can't judge
-------------------	-----------------	--------------	----------------	---------------

- *... enable my children to play outdoor*

Strongly disagree	Rather disagree	Rather agree	Strongly agree	I can't judge
-------------------	-----------------	--------------	----------------	---------------

- *... walk my dog*

Strongly disagree	Rather disagree	Rather agree	Strongly agree	I can't judge
-------------------	-----------------	--------------	----------------	---------------

- *Other reason:*

Urban watercourses and their associated greenery take many forms.

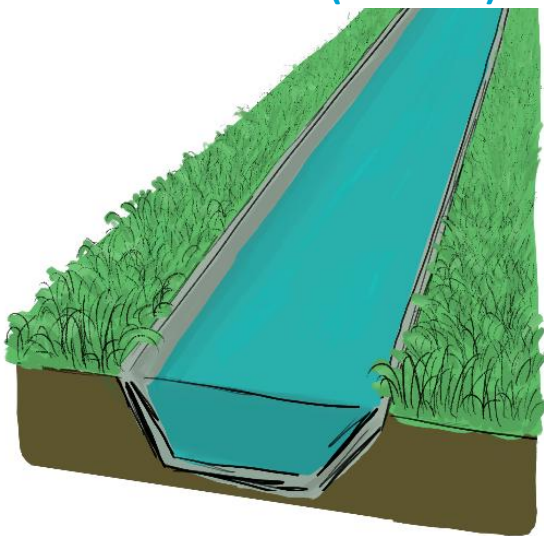
Now we will show you a number of different elements and their forms in the illustrations.

We will ask you to rate them as you like them in the city from the **AESTHETIC** point of view.

1 = least like,
10 = most like.

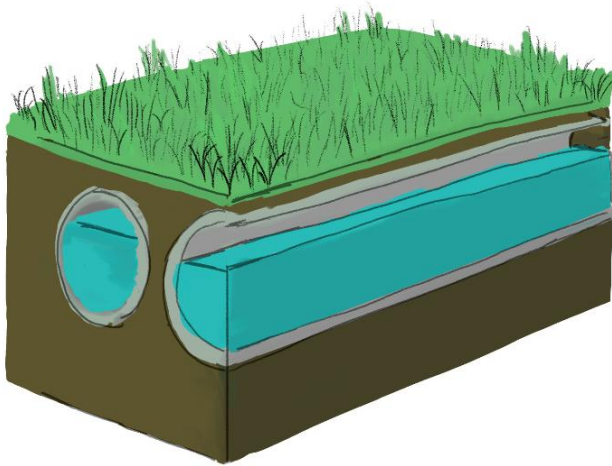
Give 1 to 10 points to the following elements:

1. Concrete river bed (corridor)



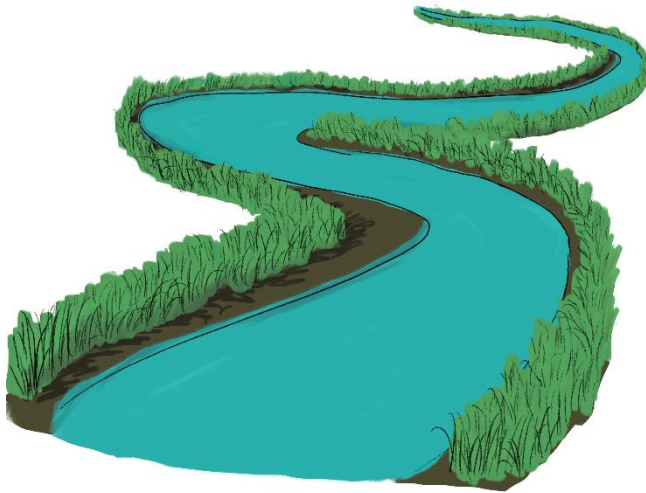
Number of points:

2. Stream forced underground in pipes



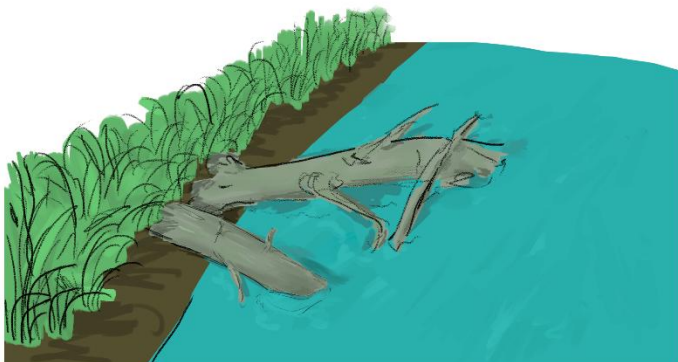
Number of points:

3. Meanders



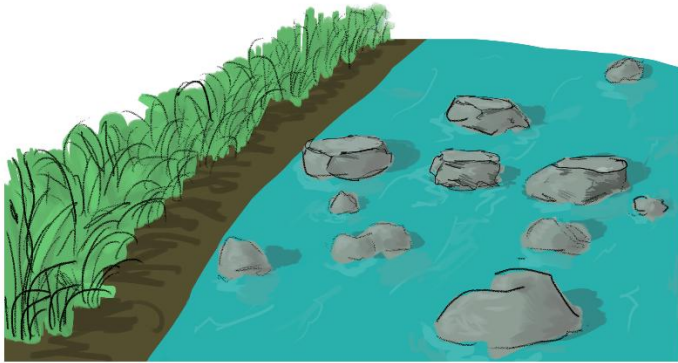
Number of points:

4. Dead wood in stream



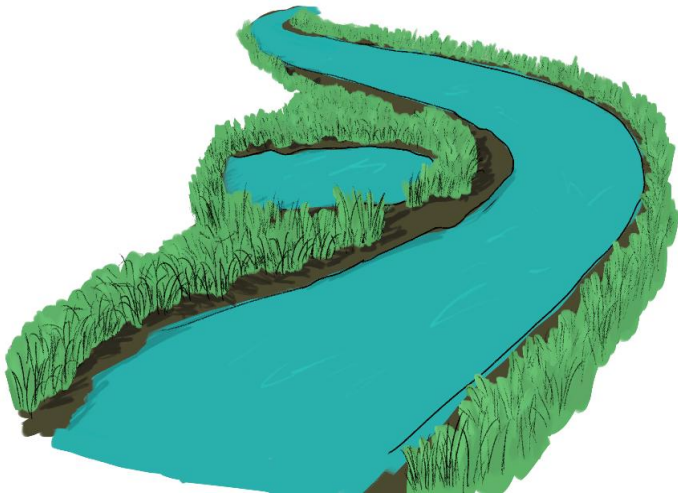
Number of points:

5. Stones in the stream



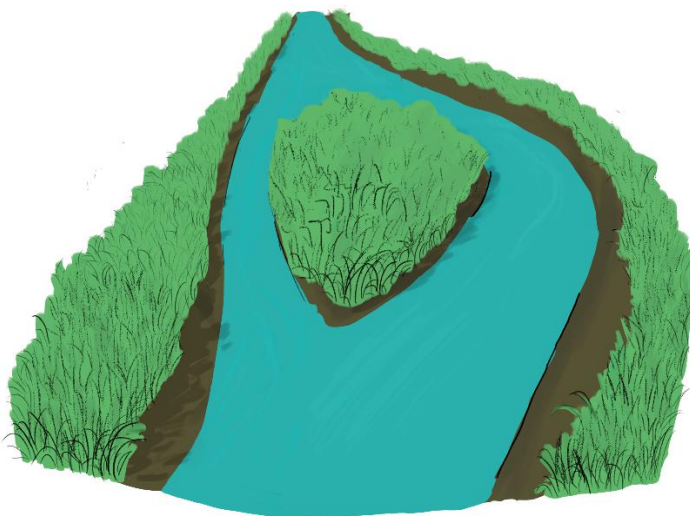
Number of points:

6. Pools and wetlands in floodplains



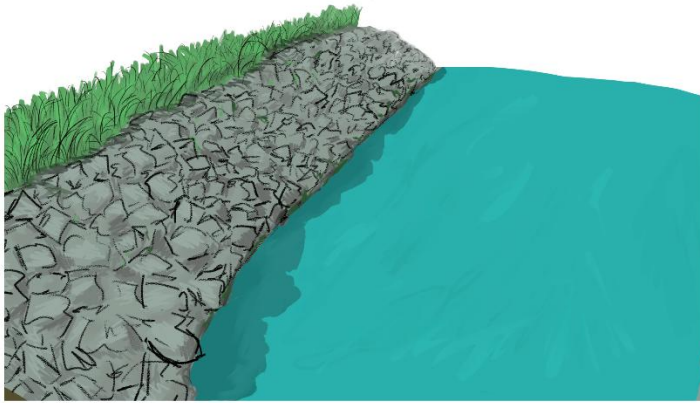
Number of points:

7. Isle in the stream



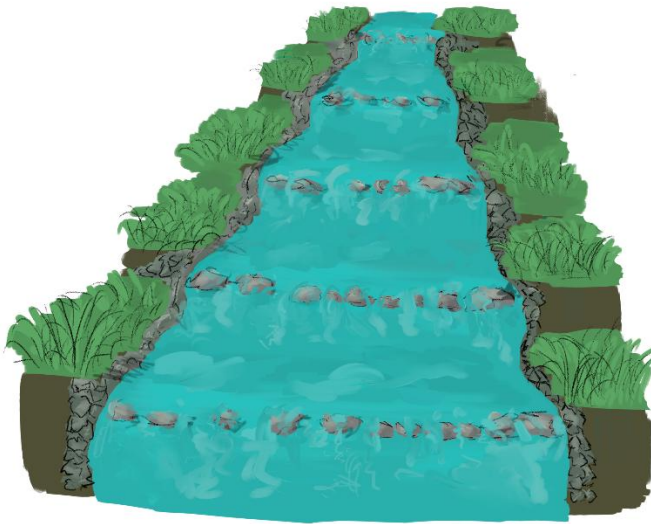
Number of points:

8. Riprap banks (rocky material placed along the stream)



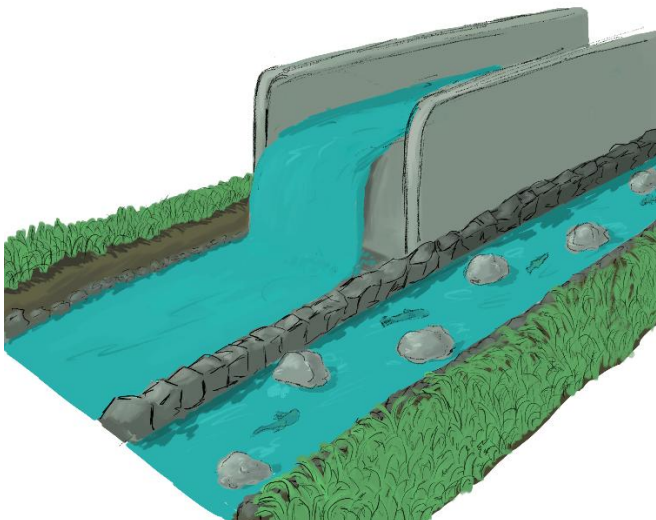
Number of points:

9. Water terraces/small dams



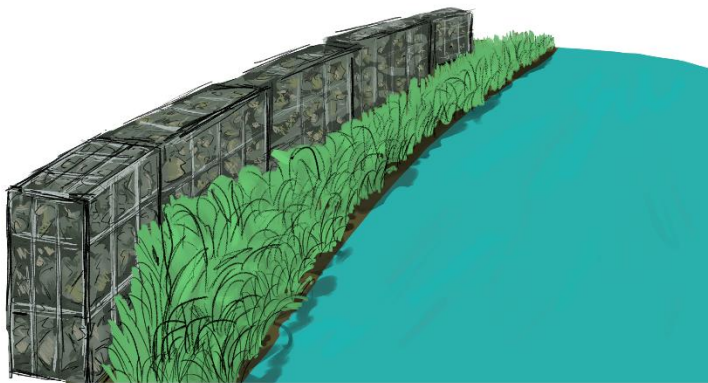
Number of points:

10. Fish passage ("Bridging the migration barriers")



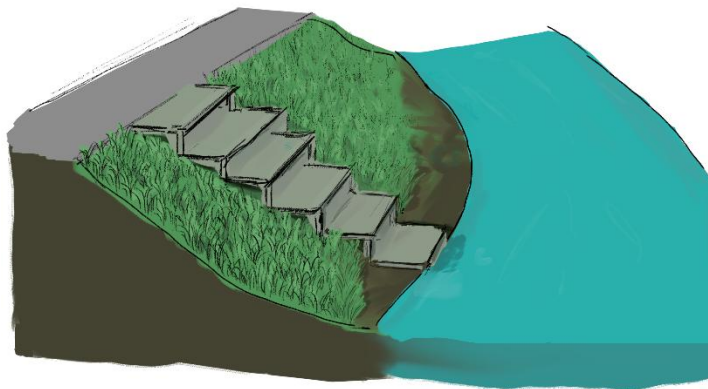
Number of points:

11. Stone flood walls



Number of points:

12. Access to the stream: Stairs



Number of points:

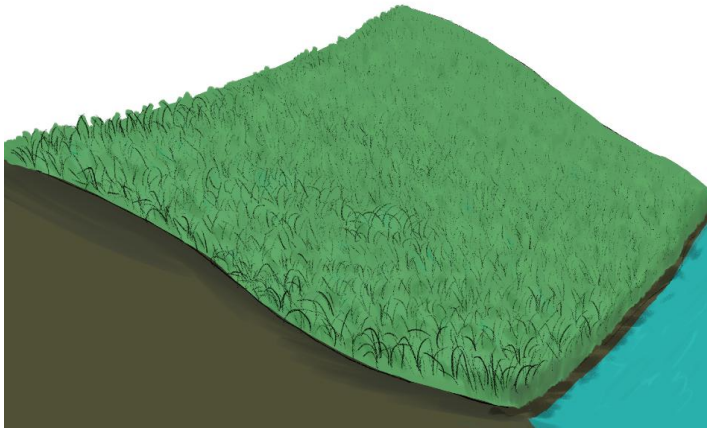
13. Access to the stream: Small pier



Number of points:

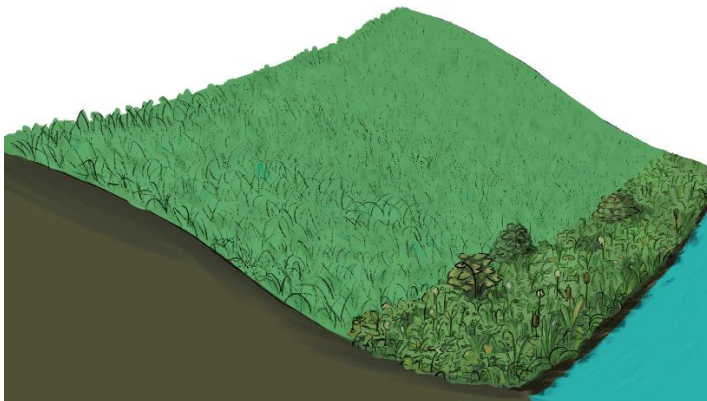
Give 1 to 10 points to the type of riparian vegetation:

1. Grassed banks



Number of points:

2. Diverse nature-based vegetation in a narrow strip along the stream, otherwise grassing



Number of points:

3. Diverse nature-based vegetation in a wide strip along the stream



Number of points:

River revitalization can provide many services. Determine on the scale, how important are these benefits to you personally?

***Completely
unimportant***

***Completely
important***

***I don't
know***

1	2	3	4	0
---	---	---	---	---

Dog walking

1	2	3	4	0
---	---	---	---	---

Recreational opportunities (e.g., play, sports, leisure activities)

1	2	3	4	0
---	---	---	---	---

Opportunities for social contacts

1	2	3	4	0
---	---	---	---	---

Education and inspiration for art and hobbies (e.g. painting)

1	2	3	4	0
---	---	---	---	---

Improvement of the microclimate (shade, cooling, humidity)

1	2	3	4	0
---	---	---	---	---

Improvement of air quality

1	2	3	4	0
---	---	---	---	---

Protection against floods

1	2	3	4	0
---	---	---	---	---

Improvement of water quality

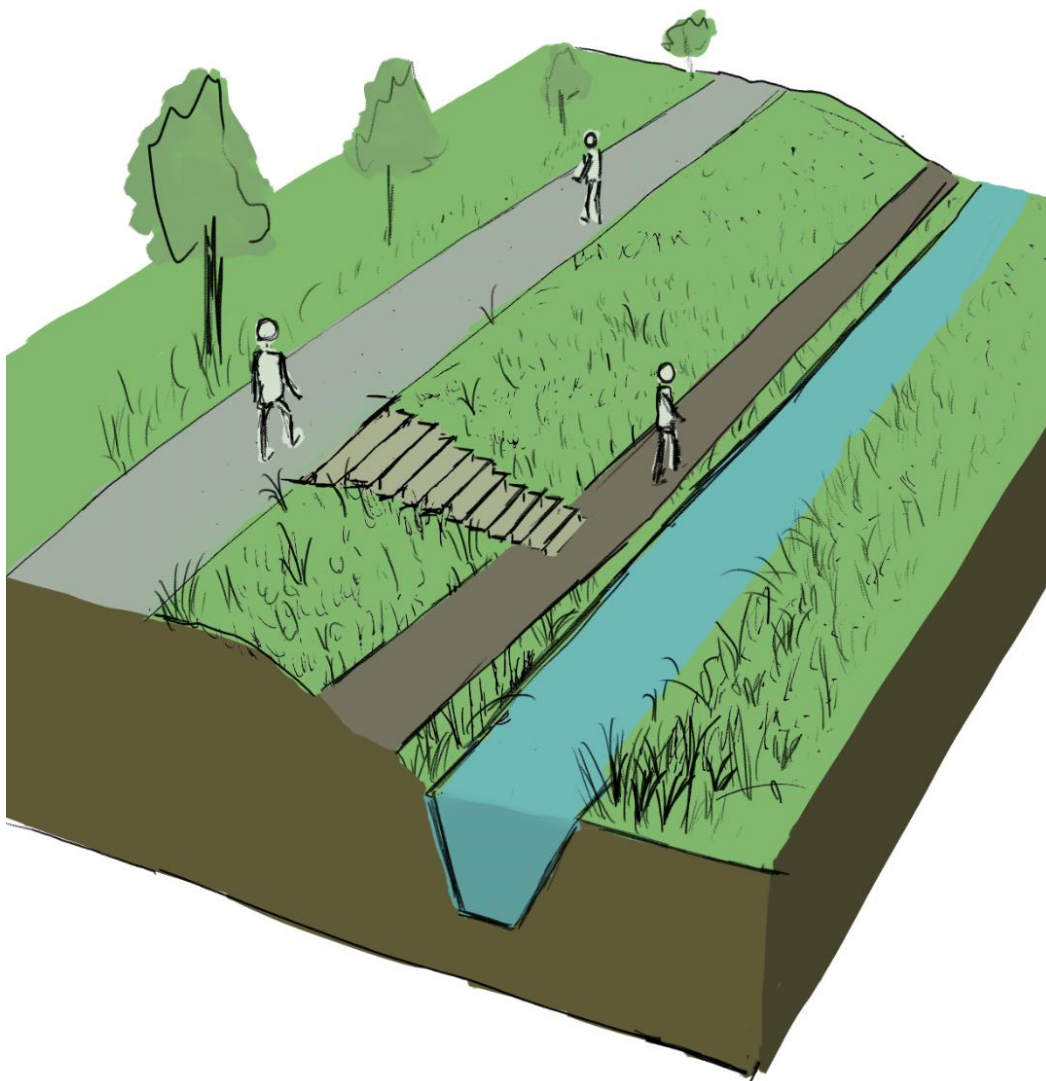
1	2	3	4	0
---	---	---	---	---

Habitat for plants and animals

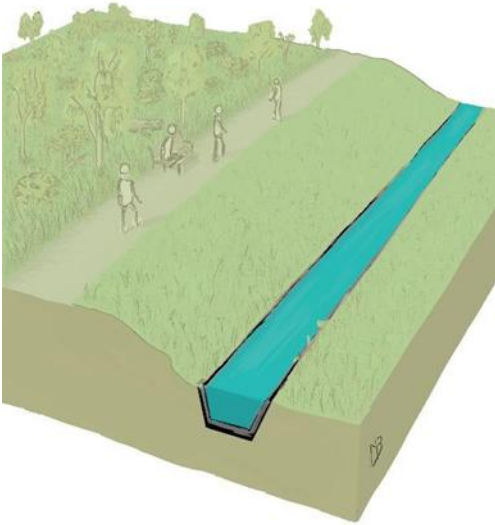
1	2	3	4	0
---	---	---	---	---

An example of an urban river course and its surroundings

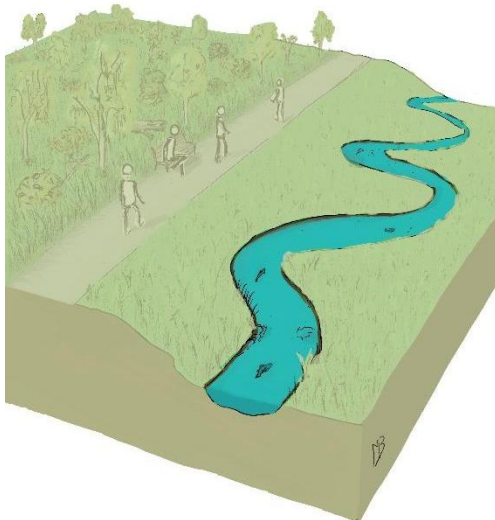
Imagine that it would be hypothetically possible to carry out the revitalization of a watercourse entirely according to your preferences (regardless of real constraints such as ownership, technical conditions, presence of utilities such as electricity, gas water, etc.). We will gradually show you different combinations of attributes, your goal will be to determine which of the combinations presented you find appropriate/preferred. The data from the preferences expressed in this way, together with the consideration of the partial barriers, will be one of the partial bases for the design and possible future revitalisation of the watercourse.



Type of river corridor



Stream in the concrete
riverbed stabilization)



Meandering stream

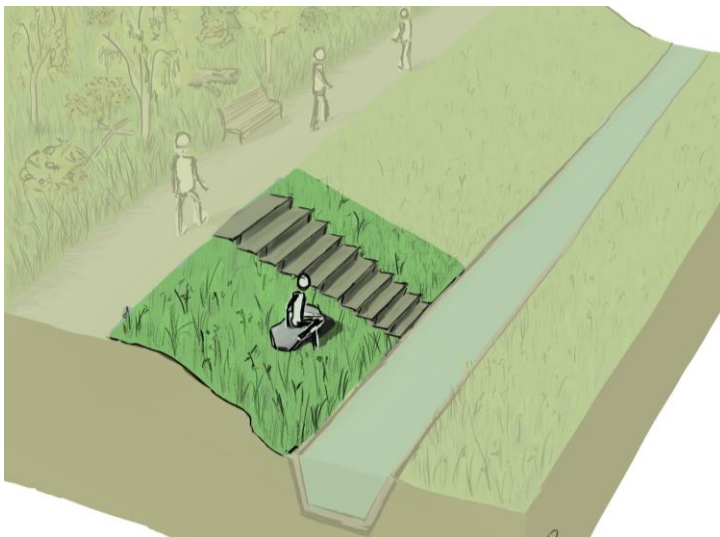


Meandering stream
and another element
(pond and island)

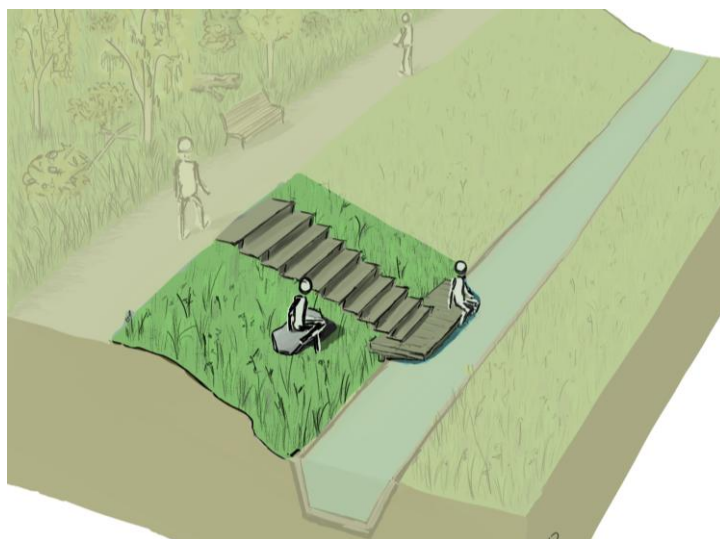
Accessibility of the water



No accessibility and no elements

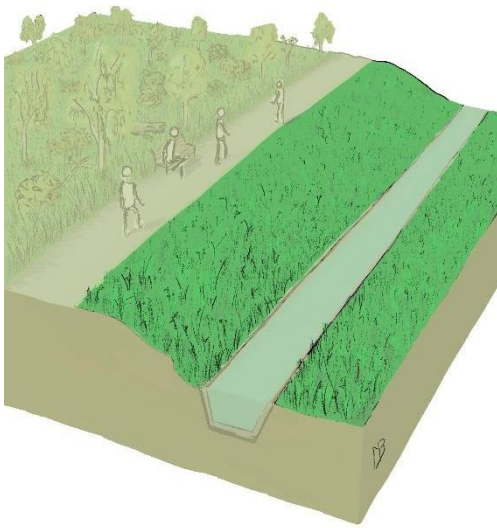


Stairs with seating possibilities near the water

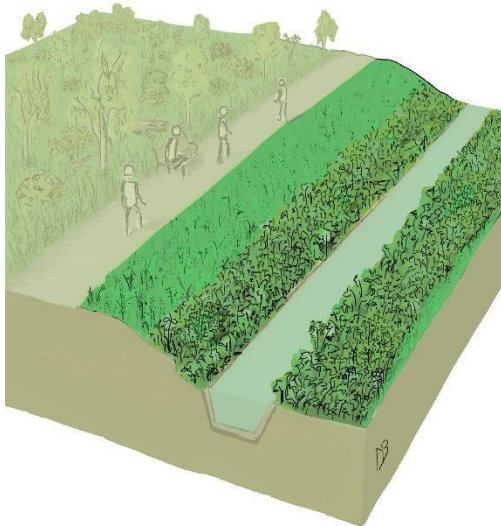


Stairs and pier at the bottom with seating possibilities near the water

Riparian vegetation



No vegetation - only
grassing



More diverse vegetation
in a narrow strip along
the stream



More diverse vegetation
in a wider strip along the
stream

Management of Greenery in wider surroundings



Nature-based

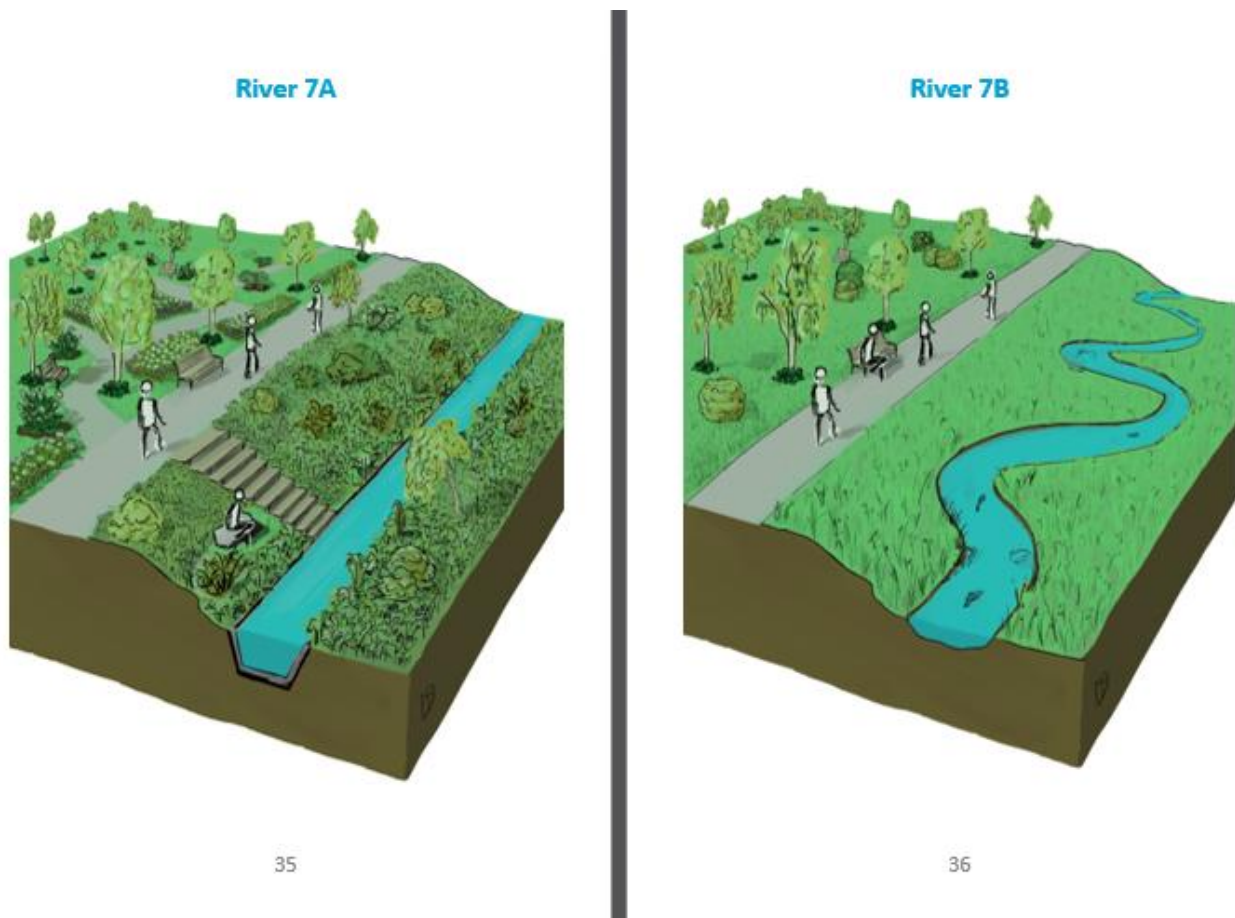


Semi-natural



Urban park

Sample of how the selection will take place



Choices:

- a) • Card (River) 1
- b) • Card (River) 2
- c) • Neither card - current status (if you are not interested in either option)

9 CHOICE EXPERIMENT CARDS

Divide the attributes into three groups according to the degree to which you considered them in your choices.

- Type of river corridor
- Accessibility of the water
- Riparian vegetation
- Management of greenery in wider surroundings

Always taken into account

Only occasionally taken i.a.

Didn't take i.a. at all

What part of city do you live in?



Which part of the city do you live?

- ☐ Area 1 (Name)
- ☐ Area 2 (Name)
- ☐ Area 3 (Name)
- ☐ Area 4 (Name)
- ☐ Area 5 (West part)
- ☐ Area 6 (Nord part)
- ☐ I don't live in this city

How much time do you usually spend to reach the area in your preferable way (walking, cycling, car, public transport or combination):

- ☐ till 5 minutes
- ☐ 6-15 minutes
- ☐ I need more time

Have you lived where you live since you were born?

- ☐ Yes
- ☐ No

For the following two statements **about the place where you currently live** (where you spend most nights), indicate how much you agree or disagree.

- **I like the place where I live now as a place to live.**

Strongly disagree	Rather disagree	Rather agree	Strongly agree	I can't judge
-------------------	-----------------	--------------	----------------	---------------

- **I plan to still be living in the place I'm living in 5 years from now.**

Strongly disagree	Rather disagree	Rather agree	Strongly agree	I can't judge
-------------------	-----------------	--------------	----------------	---------------

Where do you live?

- ☐ In a flat
- ☐ In a family house

How old are you?

How many adults (over 18), including you, live in your household?

How many children (under 18) live in your household:

What is your highest completed level of education?

- ☐ Primary
- ☐ Secondary without a diploma
- ☐ Secondary with a diploma
- ☐ Higher education

What is your employment status?

- ☐ full-time employee
- ☐ part-time employee
- ☐ private individual, entrepreneur
- ☐ unemployed
- ☐ student, apprentice
- ☐ pensioner
- ☐ permanently in the household
- ☐ on maternity/parental leave
- ☐ working student
- ☐ working pensioner

Gender

- ☐ Female
- ☐ Male
- ☐ Other

The questionnaire survey was prepared within the framework of the ReBioClim project: REstoring urban streams to promote BIOdiversity, CLIMate adaptation and to improve quality of life in cities (CE0200754), funded by the Interreg Central Europe programme.

Authors: Ing. Jiří Louda, Ph.D.; Bc. Štěpánka Truhlářová; Ing. Jan Macháč, Ph.D.; Ing. Jan Brabec, Ph.D.; Ing. Lenka Dubová, Ph.D.; et al.

Contact: jiri.louda@ujep.cz

Illustration: Matouš Bělohoubek

Web page: www.ieep.cz;
www.interreg-central.eu/projects/rebioclim/